

PRACTICE WORKSHOP

Workshop 1: Getting Started with Java

This workshop gets you comfortable with the Java toolchain and the basic building blocks of the language: variables, primitive types, operators, and console I/O.

Learning Objectives

- Set up a JDK and confirm it runs from the command line or an IDE.
- Declare and use variables of different primitive types correctly.
- Read input from the console and print formatted output.
- Use arithmetic, comparison, and logical operators in simple expressions.

Exercises

Exercise 1: Hello, You

Write a program that asks the user for their name and their birth year, then prints a greeting that includes their approximate current age.

Hints:

- Use Scanner to read input.
- Assume the current year is a constant you define at the top of the program.

```
import java.util.Scanner;

public class HelloYou {
    public static void main(String[] args) {
        // TODO: read name and birth year, then print a greeting with age
    }
}
```

Exercise 2: Temperature Converter

Write a program that converts a Celsius temperature entered by the user into Fahrenheit and Kelvin, printing both results with two decimal places.

Hints:

- Fahrenheit = Celsius * 9/5 + 32
- Kelvin = Celsius + 273.15
- Use String.format or printf for decimal formatting.

Exercise 3: Simple Grade Calculator

Ask the user for three test scores (0-100) and compute the average. Print whether the average corresponds to a Pass (≥ 50) or Fail.

Hints:

- Use an if/else statement.
- Watch out for integer division when averaging.

Stretch Goal (optional)

Extend the Grade Calculator to accept an arbitrary number of scores using a loop that stops when the user enters a negative number.

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